



Methods of Differentiation : RACE-01

Single Correct Question

1. If $y = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$, then $\frac{dy}{dx} =$

- (A) y (B) $y - 1$ (C) $y + 1$ (D) None of these

MDF153

2. $\frac{d}{dx}(e^x \log \sin 2x) =$

- (A) $e^x (\log \sin 2x + 2 \cot 2x)$
 (B) $e^x (\log \cos 2x + 2 \cot 2x)$
 (C) $e^x (\log \cos 2x + \cot 2x)$
 (D) None of these

MDF154

3. $\frac{d}{dx} \left[\left(\frac{\tan^2 2x - \tan^2 x}{1 - \tan^2 2x \tan^2 x} \right) \cot 3x \right]$

- (A) $\tan 2x \tan x$ (B) $\tan 3x \tan x$ (C) $\sec^2 x$ (D) $\sec x \tan x$

MDF155

4. If $f(x) = |\cos x|$, then $f'\left(\frac{3\pi}{4}\right)$ is equal to -

- (A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) -1

MDF156

Subjective Type Question

5. Differentiate each of following functions by first principle :

- (i) $f(x) = \tan x$ (ii) $f(x) = e^{\sin x}$

MDF157

Differentiate the given functions with respect to x :

6. $y = \sqrt{1 + \tan\left(x + \frac{1}{x}\right)}$

MDF158

7. $y = x^2 \log_3 x$

MDF159

8. $y = \log_2 [\log_3 (\log_5 x)]$

MDF160

9. $y = Ae^{-k^2x} \sin(\omega x + \alpha)$

MDF161

10. If $y = e^x \tan x + x \log_e x$, find $\frac{dy}{dx}$.

MDF162

ANSWER KEY

Question	1	2	3	4	
Answer	A	A	C	B	

6. $\frac{x^2 - 1}{2x^2 \cos^2 \left(x + \frac{1}{x} \right) \sqrt{1 + \tan \left(x + \frac{1}{x} \right)}}$

7. $2x \log_3 x + \frac{x}{\ln 3}$

8. $\frac{1}{x \log_5 x \log_3 (\log_5 x) \ln 2 \ln 3 \ln 5}$

9. $Ae^{-k^2x} [\omega \cos(\omega x + \alpha) - k^2 \sin(\omega x + \alpha)]$



Methods of Differentiation : RACE-02

Single Correct Question

1. If $y = e^{x+e^{x+e^{x+\dots\infty}}}$, then $\frac{dy}{dx} =$

- (A) $\frac{y}{1-y}$ (B) $\frac{1}{1-y}$ (C) $\frac{y}{1+y}$ (D) $\frac{y}{y-1}$

MDF163

2. $\frac{d}{dx} \{(\sin x)^x\} =$

- (A) $\left[\frac{x \cos x + \sin x \log \sin x}{\sin x} \right]$ (B) $(\sin x)^x \left[\frac{x \cos x + \sin x \log \sin x}{\sin x} \right]$
(C) $(\sin x)^x \left[\frac{x \sin x + \sin x \log \sin x}{\sin x} \right]$ (D) None of these

MDF164

3. If $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$, then $\frac{dy}{dx} =$

- (A) $\frac{-y}{x}$ (B) $\frac{y}{x}$ (C) $\frac{-x}{y}$ (D) $\frac{x}{y}$

MDF165

4. If $y = \frac{1}{4}u^4$, $u = \frac{2}{3}x^3 + 5$, then $\frac{dy}{dx} =$

- (A) $\frac{1}{27}x^2(2x^3 + 15)^3$ (B) $\frac{2}{27}x(2x^3 + 5)^3$
(C) $\frac{2}{27}x^2(2x^3 + 15)^3$ (D) None of these

MDF166

5. If $x = a(t - \sin t)$, $y = a(1 + \cos t)$, then $\frac{dy}{dx}$ equals-

- (A) $-\tan \frac{t}{2}$ (B) $\cot \frac{t}{2}$ (C) $-\cot \frac{t}{2}$ (D) $\tan \frac{t}{2}$

MDF167

Integer Type Question

6. Let $f(x) = (x-1)^{100} (x-2)^{2(99)} (x-3)^{3(98)} \dots (x-100)^{100}$, then the value of $\frac{f'(101)}{f(101)}$ is

MDF168

Subjective Type Question

Differentiate the given functions with respect to x :

7. $y = \frac{(x+1)^3 \sqrt[4]{x-2}}{\sqrt[5]{(x-3)^2}}$

MDF169

8. $y = (x^2 + 1)^{\sin x}$

MDF170

ANSWER KEY

Question	1	2	3	4	5	6	
Answer	A	B	C	C	C	5050	

7. $\frac{57x^2 - 302x + 361}{20(x-2)(x-3)} \cdot \frac{(x+1)^2 \sqrt[4]{x-2}}{\sqrt[5]{(x-3)^2}}$

8. $(x^2 + 1)^{\sin x} \left[\frac{2x \sin x}{x^2 + 1} + \cos x \ln(x^2 + 1) \right]$



Methods of Differentiation : RACE-03

Single Correct Question

1. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then $\frac{dy}{dx} =$
 (A) $1+x$ (B) $(1+x)^{-2}$ (C) $-(1+x)^{-1}$ (D) $-(1+x)^{-2}$ MDF171
2. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to -
 (A) $\frac{2^x + 2^y}{2^x - 2^y}$ (B) $\frac{2^x + 2^y}{1 + 2^{x+y}}$ (C) $2^{x-y} \left(\frac{2^y - 1}{1 - 2^x} \right)$ (D) $\frac{2^x + y - 2^x}{2^y}$ MDF172
3. If $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \dots \infty}}}$, then $\frac{dy}{dx} =$
 (A) $\frac{x}{2y-1}$ (B) $\frac{x}{2y+1}$ (C) $\frac{1}{x(2y-1)}$ (D) $\frac{1}{x(1-2y)}$ MDF173
4. If $x^3 - y^3 + 3xy^2 - 3x^2y + 1 = 0$, then at $(0, 1)$ $\frac{dy}{dx}$ equals-
 (A) 1 (B) -1 (C) 2 (D) 0 MDF174
5. If g is inverse of f and $f(x) = 2x + \sin x$; then $g'(x)$ equals :
 (A) $-\frac{3}{x^2} + \frac{1}{\sqrt{1-x^2}}$ (B) $2 + \sin^{-1}x$ (C) $2 + \cos g(x)$ (D) $\frac{1}{2 + \cos(g(x))}$ MDF175
6. If $y \sec x + \tan x + x^2 y = 0$, then $\frac{dy}{dx} =$
 (A) $\frac{2xy + \sec^2 x + y \sec x \tan x}{x^2 + \sec x}$ (B) $-\frac{(2xy + \sec^2 x + \sec x \tan x)}{x^2 + \sec x}$
 (C) $-\frac{(2xy + \sec^2 x + y \sec x \tan x)}{x^2 + \sec x}$ (D) None of these MDF176
7. If g is the inverse of f and $f'(x) = \frac{1}{1+x^3}$ then $g'(x)$ is equal to -
 (A) $1 + [g(x)]^3$ (B) $\frac{-1}{2(1+x^2)}$ (C) $\frac{1}{2(1+x^2)}$ (D) $\frac{1}{1+[g(x)]^3}$ MDF177

Integer Type Question

8. Let $g(x) = f^{-1}(x)$ where $f(x) = x + e^x$ then find $2g'(1)$.

MDF178

ANSWER KEY

Question	1	2	3	4	5	6	7	8	
Answer	D	C	C	A	D	D	A	1	



Methods of Differentiation : RACE-04

Single Correct Question

1. If $y = (A + Bx)e^{mx} + (m - 1)^{-2} e^x$ then $\frac{d^2y}{dx^2} - 2m \frac{dy}{dx} + m^2y$ is equal to -
 (A) e^x (B) e^{mx} (C) e^{-mx} (D) $e^{(1-m)x}$

MDF179

2. $\frac{d}{dx} \left(\tan^{-1} \sqrt{\frac{1 + \cos \frac{x}{2}}{1 - \cos \frac{x}{2}}} \right), x \in \left(\frac{\pi}{2}, \pi \right)$ is equal to
 (A) $-\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{4}$

MDF180

3. If $y = \tan^{-1} \frac{4x}{1+5x^2} + \tan^{-1} \frac{2+3x}{3-2x}, x > 0, x \neq \frac{3}{2}$ then $\frac{dy}{dx} =$
 (A) $\frac{1}{1+25x^2} + \frac{2}{1+x^2}$ (B) $\frac{5}{1+25x^2} + \frac{2}{1+x^2}$ (C) $\frac{5}{1+25x^2}$ (D) $\frac{1}{1+25x^2}$

MDF181

4. If $y = \tan^{-1} \left[\frac{\sin x + \cos x}{\cos x - \sin x} \right], x \in \left(0, \frac{\pi}{4} \right)$ then $\frac{dy}{dx}$ is
 (A) $\frac{1}{2}$ (B) $\frac{\pi}{4}$ (C) 0 (D) 1

MDF182

Subjective Type Question

5. If $y = xe^{x^2}$ then find y'' .

MDF183

6. Find y'' at $x = \pi/4$, if $y = x \tan x$.

MDF184

7. If $f(x) = \cos^{-1}(4x^3 - 3x)$, then find :

(a) $f' \left(-\frac{\sqrt{3}}{2} \right),$ (b) $f'(0),$ (c) $f' \left(\frac{\sqrt{3}}{2} \right)$

MDF185

8. Prove that the function $y = e^x \sin x$ satisfies the relationship $y'' - 2y' + 2y = 0$.

MDF186

ANSWER KEY

Question	1	2	3	4	
Answer	A	A	C	D	

5. $(y'' = 4y + 2xy')$ 6. $(\pi + 4)$ 7. (a) -6 (b) 3 (c) -6



Methods of Differentiation : RACE-05

Single Correct Question

1. If f is differentiable in $(0, 6)$ & $f'(4) = 5$, then $\lim_{x \rightarrow 2} \frac{f(4) - f(x^2)}{2 - x} =$
 (A) 5 (B) $5/4$ (C) 10 (D) 20 MDF187
2. Let $f(x)$ be differentiable at $x = h$, then $\lim_{x \rightarrow h} \frac{(x+h)f(x) - 2hf(h)}{x-h}$ is equal to -
 (A) $f(h) + 2hf'(h)$ (B) $2f(h) + hf'(h)$ (C) $hf(h) + 2f'(h)$ (D) $hf(h) - 2f'(h)$ MDF188
3. If $f_n(x), g_n(x), h_n(x), n = 1, 2, 3$ are polynomials in x such that $f_n(a) = g_n(a) = h_n(a), n = 1, 2, 3$ and

$$F(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$$
 Then $F'(a)$ is equal to
 (A) 0 (B) $f_1(a) g_2(a) h_3(a)$ (C) 1 (D) None of these MDF189
4. $f(x) = \begin{vmatrix} x^3 & x^2 & 3x^2 \\ 1 & -6 & 4 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant, then $\frac{d^3 f(x)}{dx^3}$ is
 (A) Proportional to x^2 (B) Proportional to x (C) Proportional to x^3 (D) A constant MDF190
5. If $y = \sin px$ and y_n is the n^{th} derivative of y , then $\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix}$ is equal to
 (A) 1 (B) 0 (C) -1 (D) None of these MDF191

Integer Type Question

6. Evaluate the following limits using L'Hospital's Rule or otherwise :

$$\lim_{x \rightarrow 0} \frac{x^{6000} - (\sin x)^{6000}}{x^2 \cdot (\sin x)^{6000}}$$
MDF192
7. If $\lim_{x \rightarrow 0} \frac{1 - \cos x \cdot \cos 2x \cdot \cos 3x \dots \cos nx}{x^2}$ has the value equal to 253, find the value of n (where $n \in \mathbb{N}$).
MDF193

8. If $f(x) = \begin{vmatrix} 2x & x^2 & x^3 \\ x^2 + 2x & 1 & 3x + 1 \\ 2x & 1 - 3x^2 & 5x \end{vmatrix}$, then find $f'(1)$.

MDF194

Subjective Type Question

9. Using L'Hospital rule find

(a) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ (b) $\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$

MDF195

10. Using L'Hospital rule verify that :

(a) $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3} = -\frac{1}{2}$ (b) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$

MDF196

11. Given a real valued function $f(x)$ as follows :

$f(x) = \frac{x^2 + 2\cos x - 2}{x^4}$ for $x < 0$; $f(0) = \frac{1}{12}$ & $f(x) = \frac{\sin x - \ln(e^x \cos x)}{6x^2}$ for $x > 0$. Test the continuity and differentiability of $f(x)$ at $x = 0$.

MDF197

12. If $f(x) = \begin{vmatrix} e^x & x^2 \\ \ln x & \sin x \end{vmatrix}$, then find $f'(1)$.

MDF198

ANSWER KEY

Question	1	2	3	4	5	6	7	8	
Answer	D	A	A	D	B	1000	11	9	

9. (a) $\frac{1}{3}$ (b) $\frac{1}{2}$

11. f is cont. but not derivable at $x = 0$.

12. $e(\sin 1 + \cos 1) - 1$